CS2470 Q

1. Define $y_i = w_i x_i + b_i$

$$L(y, a) = (y_i - a)^2$$

$$\frac{\partial L}{\partial w_i} = 2(y_i - a) \cdot \frac{\partial L}{\partial y_i} \frac{\partial y_i}{\partial w_i} = 2(y_i - a) \cdot x_i$$

2. Essentially, calculating the loss and minimizing the loss function are actually aiming to solve a large sets of ODEs. To solve these ODEs, the typical way is to numerically solve it with different algorithms. Most of time the loss is too complex to have a theoretical solution, so numerically solve it is the only way.

3. To show SGD is unbiased when batch size is 1

$$\mathbb{E}[\nabla \hat{L}] = \sum_i \mathbb{E}[\nabla \text{loss}(f(x_i), y_i)]$$

$$= \sum \frac{1}{n} \mathbb{E}[\nabla f(x_i), y_i] = \sum_{i=1}^{n} \frac{1}{n} \nabla \text{loss}(f(x_i), y_i)$$

$$= \sum \frac{1}{n} \mathbb{E}[\nabla \text{loss}(f(x_i), y_i)] = \sum_{i=1}^{n} \frac{1}{n} \nabla \text{loss}(f(x_i), y_i)$$

$$= \frac{1}{n} \nabla \sum_{i=1}^{n} \text{loss}(f(x_i), y_i)$$

$$= \nabla L$$

Hence it is unbiased.